# Chapter 7 — Arrays

An **array** is JavaScript's ordered list: shopping-cart items, search results, rows from a database, pixels in an image, moves in a game. Arrays come with about forty built-in methods, and knowing which ones **change** the array and which ones **return a new one** is the single most useful thing to learn — it prevents a whole class of bugs, especially in React and Redux code (React is a popular library for building user interfaces; Redux is a library for managing an app's shared data, its *state*).

**What you'll learn**

1. Why arrays exist, and how to create them (literal, `new Array`, `Array.of`, `Array.from`, `fill`).
2. Reading and writing by index, `at()`, `length`, holes (sparse arrays), and arrays as objects.
3. Adding and removing at the ends: `push`, `pop`, `unshift`, `shift` — and their costs.
4. The complete methods table, split into **mutating** and **non-mutating** — including the ES2023 copying methods `toSorted`, `toReversed`, `toSpliced` and `with`.
5. Iteration methods: `forEach`, `map`, `filter`, `reduce`, `find`/`findLast`, `some`/`every`, `flat`/`flatMap`, and chaining them into pipelines.
6. Sorting correctly with comparators (numbers, strings, multiple keys), stability, and shuffling.
7. Spread, rest and array destructuring; shallow vs deep copies; array-likes; everyday recipes.
8. When to use an array vs an object (vs `Map`/`Set`), and a short introduction to typed arrays.

**Before you start.** You need functions and arrow functions (Chapter 4); callbacks (functions you pass into another function so it can call them) are used everywhere here. Loops (`for`, `for…of`) are in Chapter 3. Object spread/destructuring are in Chapter 5; `Map`, `Set` and the iterator protocol behind `for…of` are in Chapter 8.

All code is in `code/`; outputs are real, from Node 24.

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## 7.1 Why arrays?

### What it is

An **array** stores many values, in order, under one name. Each value is an **element**, and each element has a numeric position called its **index**, starting at **0**.

Think of a row of numbered lockers: the row has one name (`fruits`), each locker has a number (0, 1, 2 …), and you open a locker by its number.

### The problem it solves

Without arrays you would need one variable per item:

```
const fruit1 = 'apple', fruit2 = 'banana', fruit3 = 'cherry';   // three separate variables...
const fruits = ['apple', 'banana', 'cherry'];                   // ...or one array
```

Separate variables can't be looped over, counted, sorted or passed around as a group, and adding a fourth fruit means changing code. An array handles any number of items with the same code.

### How it works

From `code/01_array_basics.mjs`:

```
console.log(fruits, fruits.length, typeof fruits, Array.isArray(fruits), Array.isArray({ length: 0 }));
const mixed = [1, 'two', { three: 3 }, [4], null, () => 5];   // any types, can nest
console.log(mixed.length, mixed[2].three, mixed[3][0], mixed[5]());
```

Output:

```
[ 'apple', 'banana', 'cherry' ] 3 object true false
6 3 4 5
```

- An **array literal** is a comma-separated list inside square brackets `[ ]`.
- `length` is the number of elements (strictly: the highest index + 1).
- Arrays are **objects**: `typeof` says `'object'`. Use `Array.isArray(x)` to test for an array.
- JavaScript arrays can mix types and nest (an array inside an array is how you build a grid or matrix — a table of values in rows and columns). In practice, keep one type per array — it's easier to reason about and engines optimise it better.

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## 7.2 Creating arrays

| Way | Example | Result | When to use |
|---|---|---|---|
| Literal | `['red', 'green', 'blue']` | 3 elements | Almost always |
| `new Array(n)` (one number) | `new Array(3)` | length 3, **no elements** (3 empty slots) | Rarely — prefer `Array.from` |
| `new Array(a, b, …)` | `new Array('a', 'b', 'c')` | those elements | Never needed — use a literal |
| `Array.of(...items)` | `Array.of(3)` | `[3]` | When a single number must be an element, not a length |
| `Array.from(iterableOrArrayLike, mapFn?)` (iterable = anything `for…of` can loop over; array-like = see section 7.7) | `Array.from({ length: 4 }, (_, i) => i * i)` | `[0, 1, 4, 9]` | Build from strings, Sets, Maps, NodeLists (lists of page elements returned by DOM queries), `arguments`, or generate by index |
| `Array(n).fill(v)` | `new Array(3).fill(0)` | `[0, 0, 0]` | Pre-filled with a **primitive** |
| Spread | `[...set]`, `[...'abc']` | copy of any iterable | Copy or convert |
| `str.split(sep)` | `'a,b'.split(',')` | `['a', 'b']` | From text |

```
console.log(new Array(3), new Array(3).length, new Array('a', 'b', 'c'), new Array(3, 4));
console.log(Array.of(3), Array.of(1, 2, 3));
console.log(Array.from('hey'), Array.from({ length: 4 }, (_, i) => i * i), Array.from(new Set([1, 1, 2])));
console.log(new Array(3).fill(0), Array(2).fill().map(() => []));
```

Output:

```
[ <3 empty items> ] 3 [ 'a', 'b', 'c' ] [ 3, 4 ]
[ 3 ] [ 1, 2, 3 ]
[ 'h', 'e', 'y' ] [ 0, 1, 4, 9 ] [ 1, 2 ]
[ 0, 0, 0 ] [ [], [] ]
```

The `new Array` constructor (a function called with `new` to build an object) is inconsistent: **one** numeric argument means "length", but two arguments mean "these elements". And `new Array(5)` does not contain five `undefined` values — it contains five **empty slots** (holes; section 7.3). `Array.of` and `Array.from` were added in ES2015 to avoid both surprises.

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## 7.3 Reading, changing, adding and removing

### Index access, `at()` and `length`

```
const colors = ['red', 'green', 'blue'];
console.log(colors[0], colors[colors.length - 1], colors.at(-1), colors[5]);
colors[2] = 'purple';
console.log(colors);
```

Output:

```
red blue blue undefined
[ 'red', 'green', 'purple' ]
```

- `colors[0]` is the first element; the last is `colors[colors.length - 1]` or, more readably, `colors.at(-1)` (ES2022 — negative indexes count from the end).
- Reading an index that doesn't exist gives `undefined` — not an error.
- Assigning to an existing index replaces that element; the array itself is changed in place.

### The four end-methods

```
const nums = [1, 2, 3];
console.log(nums.push(4), nums.unshift(0), nums);          // both return the NEW length
console.log(nums.pop(), nums.shift(), nums);               // both return the REMOVED element
```

Output:

```
4 5 [ 0, 1, 2, 3, 4 ]
4 0 [ 1, 2, 3 ]
```

| Method | Where | Returns | Cost |
|---|---|---|---|
| `push(...items)` | adds at the **end** | new length | O(1) amortised (fast on average; the occasional slow resize is spread over many calls) |
| `pop()` | removes from the **end** | the removed item (`undefined` if empty) | O(1) |
| `unshift(...items)` | adds at the **start** | new length | **O(n)** — every element shifts up one index |
| `shift()` | removes from the **start** | the removed item | **O(n)** |

(O(1) means the time doesn't grow with the array size; O(n) means it grows in proportion to the number of elements. See Chapter 17 for complexity basics.) So `push`/`pop` make a fast **stack** (last in, first out — like a pile of plates); using `shift` as a **queue** (first in, first out — like a line at a shop) on huge arrays is slow — for big queues use an index pointer or a linked structure (items that each point to the next, so removing the first never moves the others).

### `length` is writable, and holes

```
const letters = ['a', 'b', 'c', 'd'];
letters.length = 2;
console.log(letters);
letters[5] = 'f';                                          // writing past the end creates holes
console.log(letters, letters.length, 3 in letters, letters[3]);

const holey = [1, , 3];
const withUndef = [1, undefined, 3];
let visited = 0; holey.forEach(() => visited++);
console.log(holey, visited, withUndef.map((x) => x ?? 'U'), holey.map((x) => x ?? 'U'));
```

Output:

```
[ 'a', 'b' ]
[ 'a', 'b', <3 empty items>, 'f' ] 6 false undefined
[ 1, <1 empty item>, 3 ] 2 [ 1, 'U', 3 ] [ 1, <1 empty item>, 3 ]
```

- Setting `length` smaller **deletes** elements; `arr.length = 0` empties the array in place (every reference sees the empty array).
- Writing beyond the end makes a **sparse array**: the missing positions are **holes** (`3 in letters` is `false`). Reading a hole gives `undefined`, but many methods (`forEach`, `map`, `filter`, `reduce`, `some`, `every`) **skip** holes, while newer ones (`for…of`, `find`, `includes`, `Array.from`, `toSorted`, `with`, …) treat them as `undefined`. Avoid holes — they are confusing and make engines de-optimise the array (switch it to a slower general-purpose storage mode).

### Looping over an array

```
for (let i = 0; i < colors.length; i++) process.stdout.write(`${i}:${colors[i]} `);
for (const c of colors) process.stdout.write(c + ' ');
colors.forEach((c, i) => process.stdout.write(`[${i}]${c} `));
for (const [i, c] of colors.entries()) process.stdout.write(`${i}=${c} `);
```

Output (one line per loop):

```
0:red 1:green 2:purple
red green purple
[0]red [1]green [2]purple
0=red 1=green 2=purple
```

| Loop | Index? | `break` / `continue` ? | `await` inside works sequentially? | Notes |
|---|---|---|---|---|
| Classic `for (let i…)` | Yes | Yes | Yes | Most control (step by 2, go backwards) |
| `for (const x of arr)` | Via `arr.entries()` | Yes | Yes | Cleanest default |
| `arr.forEach(fn)` | Yes (2nd arg) | **No** | **No** (async callbacks aren't awaited — `await` pauses until a result that arrives later is ready, Chapter 12) | Fine for simple side effects (actions like printing, rather than computing a value) |
| `for (const i in arr)` | Yes, as **strings** | Yes | Yes | **Avoid** for arrays: iterates keys, includes inherited enumerable ones (loop-visible properties added to `Array.prototype`) |

### Arrays are references

```
const a = [1, 2]; const b = a; b.push(3);
console.log(typeof a, a, a === b, [1, 2] === [1, 2]);
```

Output:

```
object [ 1, 2, 3 ] true false
```

Like objects (Chapter 5), assigning an array copies the reference; `===` compares identity, not contents; and `const` stops re-assignment but not `push`.

### Self-check

1. `[1, 2, 3].push(4)` returns? → `4` (the new length).
2. Last element without `length - 1`? → `arr.at(-1)`.
3. `new Array(2)` vs `Array.of(2)`? → two empty slots vs `[2]`.
4. Why avoid `for…in` on arrays? → It yields string keys, also visits extra and inherited enumerable properties, and isn't meant for ordered iteration — use `for…of`.

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## 7.4 The methods table: mutating vs non-mutating

### Why it matters

A **mutating** method changes the array it's called on. That is fine for a local array you own — but if the array is shared (passed into a function, stored in React state, cached), mutation changes it for **everybody** holding a reference. React compares state (the data a component displays) by reference, so mutating an array in place means React may not re-render (redraw that part of the screen) at all. ES2023 added **copying** versions of the classic mutators so you rarely need the old "copy then mutate" dance.

### Complete reference

`length` is the only instance **property** you'll use; everything else is a method. ✎ = mutates the original.

| Method | ✎ | What it does / returns | Added |
|---|---|---|---|
| `push(...x)` / `pop()` | ✎ | Add to / remove from end; new length / removed item | ES1 |
| `unshift(...x)` / `shift()` | ✎ | Add to / remove from start | ES1 |
| `splice(start, deleteCount?, ...items)` | ✎ | Remove and/or insert anywhere; returns **removed** items | ES1 |
| `sort(compareFn?)` | ✎ | Sorts in place; returns the same array | ES1 |
| `reverse()` | ✎ | Reverses in place; returns the same array | ES1 |
| `fill(value, start?, end?)` | ✎ | Overwrites a range with one value | ES2015 |
| `copyWithin(target, start, end?)` | ✎ | Copies a range to another position inside the array | ES2015 |
| `toSorted(compareFn?)` | | **New** sorted array | **ES2023** |
| `toReversed()` | | **New** reversed array | **ES2023** |
| `toSpliced(start, deleteCount?, ...items)` | | **New** array with the splice applied | **ES2023** |
| `with(index, value)` | | **New** array with one index replaced (negative OK; out of range → `RangeError`) | **ES2023** |
| `slice(start?, end?)` | | Shallow copy of a range (negatives OK) | ES1 |
| `concat(...arraysOrValues)` | | New array joining this with others (flattens one level of array args) | ES1 |
| `map(fn)` | | New array of `fn` results, same length | ES5 |
| `filter(fn)` | | New array of elements where `fn` is truthy | ES5 |
| `flat(depth = 1)` | | New array with nested arrays flattened | ES2019 |
| `flatMap(fn)` | | `map` then `flat(1)`, in one pass | ES2019 |
| `forEach(fn)` | | Calls `fn` on each element; returns `undefined` | ES5 |
| `reduce(fn, init?)` / `reduceRight(fn, init?)` | | Folds (combines step by step) to a single value, left→right / right→left | ES5 |
| `find(fn)` / `findIndex(fn)` | | First matching element / its index (`undefined` / −1) | ES2015 |
| `findLast(fn)` / `findLastIndex(fn)` | | Same, searching from the end | ES2023 |
| `some(fn)` / `every(fn)` | | Does any / do all elements pass? (short-circuit) | ES5 |
| `includes(x, from?)` | | Boolean; finds `NaN` | ES2016 |
| `indexOf(x, from?)` / `lastIndexOf(x, from?)` | | Index using `===` (cannot find `NaN`), or −1 | ES5 |
| `at(i)` | | Element at `i`; negatives from the end | ES2022 |
| `join(sep = ',')` | | String of elements; `null`/`undefined` → empty | ES1 |
| `toString()` / `toLocaleString()` | | Comma-joined string / locale-formatted | ES1 |
| `keys()` / `values()` / `entries()` | | Iterators (objects that hand out values one at a time, Chapter 8) over indexes / values / `[index, value]` pairs | ES2015 |
| **Static:** `Array.isArray(x)`, `Array.of(...)`, `Array.from(x, fn?)`, `Array.fromAsync(x)` | | Test / create arrays (`fromAsync` awaits async iterables; ES2026, already in Node 24) | ES5–ES2026 |

(You may see `toSource()` in old lists; it was a non-standard Firefox-only method and has been removed — ignore it.)

### Mutators vs copying twins in action

From `code/02_mutating_vs_copying.mjs` — a small `show(label, returnValue, original)` helper prints what the call returned, then the original array afterwards (the file also demonstrates `reverse`, `fill` and `copyWithin`):

```
let arr = ['a', 'b', 'c', 'd', 'e'];
show('splice(1, 2)', arr.splice(1, 2), arr);                  // remove 2 at index 1
show("splice(1, 0, 'X', 'Y')", arr.splice(1, 0, 'X', 'Y'), arr);  // insert, remove nothing
show("splice(-1, 1, 'Z')", arr.splice(-1, 1, 'Z'), arr);         // replace the last one

arr = [3, 1, 2];
const sorted = arr.sort();
console.log(sorted === arr, arr);

const base = [3, 1, 2];
show('toSorted()', base.toSorted(), base);
show('toReversed()', base.toReversed(), base);
show("toSpliced(1, 1, 'x')", base.toSpliced(1, 1, 'x'), base);
show('with(0, 99)', base.with(0, 99), base);
show('with(-1, 7)', base.with(-1, 7), base);
try { base.with(10, 1); } catch (e) { console.log('with(10, 1) ->', e.constructor.name); }
show('slice(1)', base.slice(1), base);
show('concat([4], 5)', base.concat([4], 5), base);
show('[...base, 4]', [...base, 4], base);
```

Output:

```
splice(1, 2)              ["b","c"]             original: ["a","d","e"]
splice(1, 0, 'X', 'Y')    []                    original: ["a","X","Y","d","e"]
splice(-1, 1, 'Z')        ["e"]                 original: ["a","X","Y","d","Z"]
true [ 1, 2, 3 ]
toSorted()                [1,2,3]               original: [3,1,2]
toReversed()              [2,1,3]               original: [3,1,2]
toSpliced(1, 1, 'x')      [3,"x",2]             original: [3,1,2]
with(0, 99)               [99,1,2]              original: [3,1,2]
with(-1, 7)               [3,1,7]               original: [3,1,2]
with(10, 1) -> RangeError
slice(1)                  [1,2]                 original: [3,1,2]
concat([4], 5)            [3,1,2,4,5]           original: [3,1,2]
[...base, 4]              [3,1,2,4]             original: [3,1,2]
```

`splice` is the do-everything mutator: *where* to start, *how many* to delete, then *what* to insert. `sort` returns **the same array** (`sorted === arr` is `true`) — a frequent source of "I sorted a copy… oh wait". The four ES2023 methods (the "change array by copy" proposal) return a fresh array and leave the original alone. They are supported in all current browsers and Node 20+.

### A real bug: a helper that mutates its argument

```
function topTwoBad(scores) { return scores.sort((a, b) => b - a).slice(0, 2); }
function topTwoGood(scores) { return scores.toSorted((a, b) => b - a).slice(0, 2); }
const s1 = [50, 90, 70];
console.log(topTwoBad(s1), 'caller now sees', s1);
const s2 = [50, 90, 70];
console.log(topTwoGood(s2), 'caller still sees', s2);
```

Output:

```
[ 90, 70 ] caller now sees [ 90, 70, 50 ]
[ 90, 70 ] caller still sees [ 50, 90, 70 ]
```

The "bad" version silently reorders the caller's data. Before ES2023 the fix was `[...scores].sort(...)` or `scores.slice().sort(...)`; today `toSorted` says it directly.

**Immutable update cheat sheet (React/Redux style)**

| Goal | Mutating | Non-mutating |
|---|---|---|
| Add to end | `arr.push(x)` | `[...arr, x]` |
| Add to start | `arr.unshift(x)` | `[x, ...arr]` |
| Insert at `i` | `arr.splice(i, 0, x)` | `arr.toSpliced(i, 0, x)` |
| Remove at `i` | `arr.splice(i, 1)` | `arr.toSpliced(i, 1)` or `arr.filter((_, j) => j !== i)` |
| Remove by id | find index + `splice` | `arr.filter((t) => t.id !== id)` |
| Replace at `i` | `arr[i] = x` | `arr.with(i, x)` |
| Update item by id | `item.done = true` | `arr.map((t) => (t.id === id ? { ...t, done: true } : t))` |
| Sort / reverse | `arr.sort(fn)` / `arr.reverse()` | `arr.toSorted(fn)` / `arr.toReversed()` |

### Common mistakes

- Using `sort`/`reverse`/`splice` on props (the inputs a React component receives from its parent) or state; use the copying twins.
- Confusing `slice` (copy a range, non-mutating) with `splice` (edit in place).
- Expecting `push` to return the array (it returns the new length), so `arr = arr.push(x)` turns `arr` into a number.
- Thinking `const arr = […]` prevents `push`; freeze it with `Object.freeze(arr)` if needed (shallow).

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## 7.5 Iteration methods: map, filter, reduce and friends

### What it is

These methods take a **callback** — a function you pass in — and call it for each element with three arguments: `(element, index, array)`. They let you describe *what* to do with each element instead of writing loop bookkeeping. This style is called **declarative** (say what you want) as opposed to **imperative** (say how to loop).

### How it works

From `code/03_iteration_methods.mjs`, using this data:

```
const orders = [
  { id: 1, customer: 'Ana', total: 120, status: 'paid' },
  { id: 2, customer: 'Ben', total: 35, status: 'pending' },
  { id: 3, customer: 'Ana', total: 60, status: 'paid' },
  { id: 4, customer: 'Cy', total: 250, status: 'refunded' },
];
```

**`forEach`**, **`map`**, **`filter`**, **`reduce`**:

```
orders.forEach((o, i) => console.log(i, o.id, o.customer));
console.log(orders.map((o) => o.total));
console.log(['1', '2', '3'].map(Number), ['1', '2', '3'].map(parseInt)); // classic trap
const paid = orders.filter((o) => o.status === 'paid');
console.log(paid.map((o) => o.id));
const revenue = paid.reduce((sum, o) => sum + o.total, 0);
const byCustomer = orders.reduce((acc, o) => { acc[o.customer] = (acc[o.customer] ?? 0) + o.total; return acc; },
{});
console.log(revenue, byCustomer);
try { [].reduce((a, b) => a + b); } catch (e) { console.log('empty reduce without initial ->',
e.constructor.name); }
console.log(['a', 'b', 'c'].reduceRight((acc, x) => acc + x, ''));
```

Output:

```
0 1 Ana
1 2 Ben
2 3 Ana
3 4 Cy
[ 120, 35, 60, 250 ]
[ 1, 2, 3 ] [ 1, NaN, NaN ]
[ 1, 3 ]
180 { Ana: 180, Ben: 35, Cy: 250 }
empty reduce without initial -> TypeError
cba
```

| Method | Callback returns | Method returns | Think of it as |
|---|---|---|---|
| `forEach` | ignored | `undefined` | "do something with each" |
| `map` | the new element | new array, **same length** | "transform each" |
| `filter` | truthy/falsy | new array, **same or fewer** elements | "keep some" |
| `reduce` | the new accumulator (the running result carried from one step to the next) | the final accumulator | "combine all into one" |

How `reduce` works, step by step, for `sum` over totals `[120, 60]` with initial value `0`:

1. `sum = 0` (the initial value), element 120 → returns 120.
2. `sum = 120`, element 60 → returns 180.
3. No elements left → `reduce` returns 180.

**Always pass the initial value.** Without it, `reduce` uses the first element as the starting accumulator — which throws on an empty array and gives wrong types when you're building an object.

**The `map(parseInt)` trap**: `map` passes `(element, index, array)`, and `parseInt` takes `(string, radix)` (radix = the number base, such as 10), so it is called as `parseInt('2', 1)` and `parseInt('3', 2)` — invalid radix and invalid digit → `NaN`. Wrap it: `map((s) => parseInt(s, 10))` or use `map(Number)`.

**Chaining into a pipeline:**

```
const report = orders
  .filter((o) => o.status !== 'refunded')
  .map((o) => ({ ...o, total: o.total * 1.1 }))
  .toSorted((a, b) => b.total - a.total)
  .map((o) => `${o.id}:${o.total.toFixed(2)}`)
  .join(', ');
console.log(report);
```

Output:

```
1:132.00, 3:66.00, 2:38.50
```

Each step returns a new array, so steps chain naturally and the original `orders` is never modified. Each step also loops over the data once; for normal UI-sized lists (hundreds or thousands) that's irrelevant, but for millions of rows a single `for` loop or lazy iterator helpers (lazy = each value is computed only when it is asked for) (Chapter 8) avoid building intermediate arrays.

**Searching:**

```
console.log(orders.find((o) => o.total > 100).id, orders.findIndex((o) => o.total > 100));
console.log(orders.findLast((o) => o.customer === 'Ana').id, orders.findLastIndex((o) => o.customer === 'Ana'));
console.log(orders.find((o) => o.total > 1000), orders.findIndex((o) => o.total > 1000));
console.log(orders.some((o) => o.status === 'refunded'), orders.every((o) => o.total > 0), [].every(() => false));
const vals = [1, 2, NaN, 2];
console.log(vals.includes(NaN), vals.indexOf(NaN), vals.indexOf(2), vals.lastIndexOf(2), vals.includes('2'));
```

Output:

```
1 0
3 2
undefined -1
true true true
true -1 1 3 false
```

| Question | Method | Not found |
|---|---|---|
| Which element matches? | `find` / `findLast` | `undefined` |
| At what index? | `findIndex` / `findLastIndex` | `-1` |
| Is this exact value present? | `includes` (handles `NaN`) | `false` |
| Where is this exact value? | `indexOf` / `lastIndexOf` (strict `===`, never finds `NaN`) | `-1` |
| Does any element pass? | `some` | `false` (also for an empty array) |
| Do all elements pass? | `every` | `true` for an empty array ("vacuous truth") |

`find`, `some` and `every` **stop early** as soon as the answer is known. Note that `includes` and `indexOf` compare with `===`-style equality, so `includes('2')` doesn't find the number 2, and searching for an *object* only works with the same reference — use `some((o) => o.id === id)` for objects.

**Flattening:**

```
const nested = [1, [2, [3, [4]]]];
console.log(nested.flat(), nested.flat(2), nested.flat(Infinity));
const sentences = ['hello world', 'good morning'];
console.log(sentences.map((s) => s.split(' ')), sentences.flatMap((s) => s.split(' ')));
console.log([1, 2, 3, 4].flatMap((n) => (n % 2 ? [] : [n, n * 10])));   // filter + map in one pass
```

Output:

```
[ 1, 2, [ 3, [ 4 ] ] ] [ 1, 2, 3, [ 4 ] ] [ 1, 2, 3, 4 ]
[ [ 'hello', 'world' ], [ 'good', 'morning' ] ] [ 'hello', 'world', 'good', 'morning' ]
[ 2, 20, 4, 40 ]
```

`flat(depth)` removes `depth` levels of nesting (default 1; `Infinity` for all). `flatMap` maps each element to an *array* and joins the results one level deep — returning `[]` drops an element, returning several items expands it, so it can filter and map in a single pass.

**Iterators, joining, and the callback signature:**

```
console.log([...['x', 'y'].keys()], [...['x', 'y'].values()], [...['x', 'y'].entries()]);
console.log([1, [2, 3], null, undefined, 4].join('-'), String([1, [2, 3]]), [1234.5, 99].toLocaleString('de-DE'));
console.log(['a', 'b'].map((el, idx, arr) => `${el}${idx}/${arr.length}`));
let seen = 0;
for (const o of orders) { seen++; if (o.total > 100) break; }
console.log('stopped after', seen);
```

Output:

```
[ 0, 1 ] [ 'x', 'y' ] [ [ 0, 'x' ], [ 1, 'y' ] ]
1-2,3---4 1,2,3 1.234,5,99
[ 'a0/2', 'b1/2' ]
stopped after 1
```

`join` turns `null`/`undefined` into empty strings and nested arrays into their comma-joined form. `forEach` has no way to stop early (a `return` inside the callback only ends that one call); use `for…of` with `break`, or `some`/`find`.

### Trade-offs: methods vs loops

| | Array methods | `for` / `for…of` loops |
|---|---|---|
| Readability | Name the intent (`filter`, `map`) | Generic; intent must be read from the body |
| Mutation risk | Encourage new arrays | Easy to mutate by accident |
| Early exit | Only `find` / `some` / `every` | `break` / `return` anywhere |
| `await` inside | Callbacks are not awaited sequentially | `for…of` + `await` runs in order |
| Performance | Slight overhead per call; intermediate arrays | Fastest; one pass |

Use methods by default; switch to a loop when you need early exit, sequential `await` (Chapter 12), or measured performance.

### Real-world examples

- Rendering lists in React: `items.map((item) => <Row key={item.id} {...item} />)` (React book).
- API data cleanup: `rows.filter(isValid).map(normalize)`.
- Totals in a shopping cart: `cart.reduce((sum, i) => sum + i.price * i.qty, 0)`.
- Feature flags (switches that turn a feature on for some users): `user.roles.some((r) => r === 'admin')`.

### In the interview

> **30-second answer:** "`map` transforms, `filter` selects, `reduce` folds to one value, `find` returns the first match, `some` / `every` answer yes/no with short-circuiting. They take `(element, index, array)` callbacks and don't mutate. I always give `reduce` an initial value and avoid `forEach` when I need `break` or sequential `await`."

- *Implement `map` with `reduce`.* `arr.reduce((acc, x, i) => { acc.push(fn(x, i, arr)); return acc; }, [])`.
- *Why does `['1','2','3'].map(parseInt)` return `[1, NaN, NaN]`?* The index is passed as the radix.
- *`find` vs `filter`?* `find` returns the first element (or `undefined`) and stops; `filter` always scans everything and returns an array.

### Self-check

1. `[1, 2, 3].map((x) => x * 2)`? → `[2, 4, 6]`.
2. `[5, 12, 8].find((x) => x > 6)`? → `12`.
3. `[].every(Boolean)`? → `true`.
4. Sum `[1, 2, 3]` with `reduce`. → `arr.reduce((s, x) => s + x, 0)`.
5. `[[1], [2, [3]]].flat()`? → `[1, 2, [3]]`.

---

## 7.6 Sorting correctly

### The problem it solves

`sort()` with no arguments converts elements to **strings** and orders them by UTF-16 code units (the 16-bit numbers JavaScript uses to store each character). That's right for simple ASCII (plain English letters and digits) words and **wrong for numbers**.

From `code/04_sorting.mjs`:

```
console.log([10, 9, 1, 100, 25].sort());                  // default: compared as STRINGS
console.log([10, 9, 1, 100, 25].sort((a, b) => a - b));   // ascending numbers
console.log([10, 9, 1, 100, 25].sort((a, b) => b - a));   // descending numbers
console.log([3, undefined, 1, , 2].sort());               // undefined and holes go last
```

Output:

```
[ 1, 10, 100, 25, 9 ]
[ 1, 9, 10, 25, 100 ]
[ 100, 25, 10, 9, 1 ]
[ 1, 2, 3, undefined, <1 empty item> ]
```

## How it works — the comparator contract

A **comparator** is a function `(a, b) => number`:

1. Return a **negative** number → `a` comes before `b`.
2. Return a **positive** number → `b` comes before `a`.
3. Return **0** → keep their current relative order.

`a - b` does this for numbers (and for Dates, since subtraction converts them to milliseconds). The comparator must be **consistent** — returning a boolean (`a > b`) is a bug: `false` becomes 0 and `true` becomes 1, never negative, so results depend on the engine's algorithm.

```
console.log('boolean comparator ->', [3, 1, 2].sort((a, b) => a > b));
```

Output:

```
boolean comparator -> [ 3, 1, 2 ]
```

### Strings, multiple keys and stability

```
const people = [
  { name: 'Zoe', dept: 'eng', age: 31 },
  { name: 'adam', dept: 'ops', age: 25 },
  { name: 'Émile', dept: 'eng', age: 25 },
  { name: 'Bob', dept: 'ops', age: 31 },
];
console.log(people.toSorted((a, b) => a.age - b.age).map((p) => p.name));
console.log(people.map((p) => p.name).toSorted());                          // code-unit order
console.log(people.map((p) => p.name).toSorted((a, b) => a.localeCompare(b)));  // human order

const byDeptAgeName = (a, b) =>
  a.dept.localeCompare(b.dept) || b.age - a.age || a.name.localeCompare(b.name);
console.log(people.toSorted(byDeptAgeName).map((p) => `${p.dept}/${p.age}/${p.name}`));

const tasks = [['write', 2], ['test', 1], ['deploy', 2], ['plan', 1]];
console.log(tasks.toSorted((a, b) => a[1] - b[1]).map((t) => t[0]));
```

Output:

```
[ 'adam', 'Émile', 'Zoe', 'Bob' ]
[ 'Bob', 'Zoe', 'adam', 'Émile' ]
[ 'adam', 'Bob', 'Émile', 'Zoe' ]
[ 'eng/31/Zoe', 'eng/25/Émile', 'ops/31/Bob', 'ops/25/adam' ]
[ 'test', 'plan', 'write', 'deploy' ]
```

- For strings, use **`localeCompare`** (or a reused `Intl.Collator().compare`) — the default order puts all capitals first and accented letters last (Chapter 6).
- **Multi-key sort**: chain comparisons with `||`. When the first comparison returns 0 (a tie), `||` falls through to the next key. `b.age - a.age` sorts that key descending.
- **Stability**: since ES2019 `sort` must be **stable** — elements that compare equal keep their original order (`test` stays before `plan`, `write` before `deploy`). That lets you sort by a secondary key first and the primary key second.

### Computed keys, booleans, dates

```
const words = ['banana', 'Kiwi', 'apple', 'fig'];
const sortedByLength = words
  .map((w) => ({ w, key: w.length }))
  .sort((a, b) => a.key - b.key || a.w.localeCompare(b.w))
  .map((x) => x.w);
console.log(sortedByLength);
console.log([true, false, true].toSorted((a, b) => Number(a) - Number(b)));
const dates = [new Date('2024-03-01'), new Date('2023-12-25'), new Date('2024-01-15')];
console.log(dates.toSorted((a, b) => a - b).map((d) => d.toISOString().slice(0, 10)));
```

Output:

```
[ 'fig', 'Kiwi', 'apple', 'banana' ]
[ false, true, true ]
[ '2023-12-25', '2024-01-15', '2024-03-01' ]
```

When the sort key is expensive to compute (parsing a date string, normalizing text), compute it **once per element** first, sort by the stored key, then strip it — the "decorate–sort–undecorate" pattern (Schwartzian transform). Otherwise the comparator recomputes keys O(n log n) times (a bit more than proportional to *n* — the usual cost of a good sort).

### Shuffling

```
function shuffle(arr, rand = Math.random) {
  const a = [...arr];
  for (let i = a.length - 1; i > 0; i--) {
    const j = Math.floor(rand() * (i + 1));
    [a[i], a[j]] = [a[j], a[i]];
  }
  return a;
}
```

This is the **Fisher–Yates** shuffle: walk from the end, swapping each position with a random earlier (or same) position — every ordering is equally likely, in O(n). The popular one-liner `arr.sort(() => Math.random() - 0.5)` is **biased** (the comparator is inconsistent) and slower. (The demo passes a seeded generator — a fake random-number source that always produces the same sequence from the same starting "seed" — so its output is reproducible: `[ 1, 2, 5, 6, 3, 4 ]`.)

### Complexity

V8 (Chrome, Node) implements `sort` with TimSort and other engines use merge-sort variants (sorting algorithms that split the list, sort the parts and merge them back): **O(n log n)** comparisons, stable, and fast on partly-sorted data.

## In the interview

**30-second answer:** "Default `sort` compares strings, so numbers need `(a, b) => a - b`. The comparator returns negative, zero or positive; chain keys with `||`; strings use `localeCompare`. `sort` mutates and is stable since ES2019; use `toSorted` to keep the original."

- *Sort objects by `lastName`, then `firstName`?* `(a, b) => a.lastName.localeCompare(b.lastName) || a.firstName.localeCompare(b.firstName)`.
- *Why not shuffle with `sort(() => Math.random() - 0.5)`?* Biased and O(n log n); use Fisher–Yates.

## Self-check

1. `[2, 10, 1].sort()`? → `[1, 10, 2]`.
2. Descending numeric comparator? → `(a, b) => b - a`.
3. What does "stable" mean? → Equal elements keep their original relative order.

---

## 7.7 Spread, rest and destructuring with arrays

From `code/05_spread_destructuring.mjs`:

```
const a = [1, 2], b = [3, 4];
console.log([...a, ...b], [0, ...a, 2.5, ...b], Math.max(...b), [...'hi'], [...new Set([1, 1, 2])]);

const [first, second, third = 'default', ...rest] = ['x', 'y'];
console.log(first, second, third, rest);
const [, , keep] = ['skip', 'skip', 'take'];
console.log(keep);
let p = 1, q = 2;
[p, q] = [q, p];
console.log(p, q);
const [[r1c1, r1c2], [r2c1]] = [[1, 2], [3, 4]];
console.log(r1c1, r1c2, r2c1);
const { 0: head, length } = ['h', 'e', 'y'];
console.log(head, length);

const minMax = (arr) => [Math.min(...arr), Math.max(...arr)];
const [lo, hi] = minMax([5, 3, 9]);
console.log(lo, hi);
const sum = (...nums) => nums.reduce((s, n) => s + n, 0);
console.log(sum(1, 2, 3), sum(...[10, 20]));
```

Output:

```
[ 1, 2, 3, 4 ] [ 0, 1, 2, 2.5, 3, 4 ] 4 [ 'h', 'i' ] [ 1, 2 ]
x y default []
take
2 1
1 2 3
h 3
6 30
```

- **Spread** `...arr` expands any **iterable** (array, string, Set, Map, generator) into elements — in array literals and in function calls.
- **Array destructuring** assigns by **position**. Defaults apply for `undefined`; commas skip positions; `...rest` gathers the remainder (always an array, possibly empty).
- `[p, q] = [q, p]` swaps two variables.
- Returning an array and destructuring it (`const [lo, hi] = minMax(...)`) is a light way to return multiple values — React's `const [state, setState] = useState(0)` (a *hook* that gives a component a remembered value plus a function to change it) uses exactly this.
- A **rest parameter** `(...nums)` collects any number of arguments into a real array (replacing the old `arguments` object — Chapter 4).

### Copies are shallow; the `fill([])` trap

```
const grid = [[0, 0], [0, 0]];
const shallow = [...grid];
const deep = structuredClone(grid);
shallow[0][0] = 9;
console.log(grid[0][0], deep[0][0]);

const bad = Array(3).fill([]);
bad[0].push('x');
const good = Array.from({ length: 3 }, () => []);
good[0].push('x');
console.log(bad, good);
```

Output:

```
9 0
[ [ 'x' ], [ 'x' ], [ 'x' ] ] [ [ 'x' ], [], [] ]
```

`[...arr]`, `slice()`, `concat()` and `Array.from(arr)` copy only the outer array; inner arrays/objects are shared. `structuredClone` copies deeply. `fill(obj)` puts the **same** object into every slot; use `Array.from({ length }, () => newThing)` to create a fresh one per slot — the standard way to build a 2-D grid.

### Array-likes

An **array-like** is an object with numeric keys and a `length` but without array methods — the old `arguments` object, DOM `NodeList`/`HTMLCollection`, strings. Convert with `Array.from` (or spread, if it's iterable), or borrow a method with `call`:

```
const arrayLike = { 0: 'a', 1: 'b', length: 2 };
console.log(Array.from(arrayLike), Array.prototype.map.call(arrayLike, (s) => s.toUpperCase()));
```

Output:

```
[ 'a', 'b' ] [ 'A', 'B' ]
```

### Everyday recipes

```
const range = (start, end, step = 1) => Array.from({ length: Math.ceil((end - start) / step) }, (_, i) => start + i * step);
const unique = (arr) => [...new Set(arr)];
const chunk = (arr, size) => Array.from({ length: Math.ceil(arr.length / size) }, (_, i) => arr.slice(i * size, i * size + size));
const zip = (xs, ys) => xs.map((x, i) => [x, ys[i]]);
const last = (arr) => arr.at(-1);
const arraysEqual = (x, y) => x.length === y.length && x.every((v, i) => Object.is(v, y[i]));
```

Output of the demo calls:

```
[ 0, 1, 2, 3, 4 ] [ 10, 7, 4, 1 ]
[ 3, 1, 2 ]
[ [ 1, 2 ], [ 3, 4 ], [ 5 ] ]
[ [ 'a', 1 ], [ 'b', 2 ] ]
3 undefined
true true
```

The last `true` is `[1, 2] == '1,2'` — loose equality converts the array to the string `'1,2'`. Never compare arrays with `==`/`===`; compare element by element (as `arraysEqual` does) or, for simple data, compare `JSON.stringify` results. Grouping arrays into objects/Maps (`Object.groupBy`) is in Chapter 8.

---

## 7.8 Arrays vs objects (vs Map/Set)

### When to pick which

| Use… | When | Example |
|---|---|---|
| **Array** | Order matters; you access by position; you iterate, sort, filter | A playlist, search results, a queue of jobs |
| **Object** | A fixed set of named fields describing one thing | `{ name, email, age }` — a record |
| **Array of objects** | A list of records — the most common shape of real data | API response `[{ id, title }, …]` |
| **Object / Map keyed by id** | Frequent lookup by a key | `usersById[id]` — O(1) instead of `find` O(n) |
| **Set** | Uniqueness and fast "is it in?" checks | Visited URLs, selected ids |

From `code/01_array_basics.mjs`:

```
const fruitList = ['Apple', 'Banana', 'Cherry'];                    // ordered, by position
const fruitDetails = {                                              // lookup by name
  apple: { color: 'red', taste: 'sweet' },
  cherry: { color: 'red', taste: 'tart' },
};
console.log(fruitList[1], fruitDetails.cherry.taste);
const fruitRecords = [                                              // most common in real apps
  { name: 'Apple', color: 'red' }, { name: 'Banana', color: 'yellow' },
];
console.log(fruitRecords.find((f) => f.name === 'Banana').color);
```

Output:

```
Banana tart
yellow
```

A simple **ordered list** of names is an array. Once each fruit carries its own **details** looked up by name, an object keyed by name reads better. Real apps usually hold an **array of objects** and, when lookups by id become frequent, build an index: `Object.fromEntries(list.map((x) => [x.id, x]))` or `new Map(list.map((x) => [x.id, x]))` (Chapter 8).

### Common mistakes

- Using `arr.includes(x)` or `arr.find(...)` inside a loop over another big array — O(n²). Build a `Set` or `Map` first.
- Using an object with numeric keys as a list — you lose `length`, order guarantees are subtler, and no array methods.
- Using an array as a dictionary (`arr['name'] = …`) — it works (arrays are objects) but `length` ignores it and methods skip it.

---

## 7.9 Typed arrays (a short note)

### What it is

A **typed array** is a fixed-length array of raw numbers of one machine type (8-bit unsigned integers, 32-bit floats, …) stored in a contiguous block of memory called an `ArrayBuffer`. Several typed-array **views** can look at the same buffer in different ways. They're used for binary data: files, images, audio, WebGL/WebGPU (browser APIs for GPU graphics), WebSockets (a two-way, always-open connection between browser and server), `fetch` bodies, WebAssembly memory (WebAssembly runs fast compiled code from languages like C or Rust in the browser), and Node's `Buffer` (a `Uint8Array` subclass — a specialised version that inherits all its behaviour).

From `code/06_typed_arrays.mjs`:

```
const buf = new ArrayBuffer(8);                 // 8 raw bytes, all zero
const bytes = new Uint8Array(buf);              // view them as 8 unsigned bytes
const words = new Uint16Array(buf);             // ...or as 4 unsigned 16-bit ints (same memory)
bytes[0] = 0x34; bytes[1] = 0x12;
console.log(bytes, words[0].toString(16), '(little-endian on this machine)');

const u8 = new Uint8Array([255, 256, -1, 3.7]);
const clamped = new Uint8ClampedArray([255, 256, -1, 3.7]);
const i8 = new Int8Array([127, 128]);
console.log(u8, clamped, i8);

const f = new Float64Array([3, 1, 2]);
console.log(f.length, typeof f.push, f.map((x) => x * 2), f.toSorted(), Array.isArray(f));
console.log(Float32Array.of(0.1)[0], new Float16Array([0.1])[0]);

console.log(Array.from(f), [...new TextEncoder().encode('Hi')], new TextDecoder().decode(new Uint8Array([72,
105])));
const view = new DataView(buf);
view.setUint16(2, 0xabcd, false);                                    // big-endian write, e.g. a network protocol
console.log(bytes.slice(2, 4), view.getUint16(2, false).toString(16));
```

Output:

```
Uint8Array(8) [
  52, 18, 0, 0,
   0,  0, 0, 0
] 1234 (little-endian on this machine)
Uint8Array(4) [ 255, 0, 255, 3 ] Uint8ClampedArray(4) [ 255, 255, 0, 4 ] Int8Array(2) [ 127, -128 ]
3 undefined Float64Array(3) [ 6, 2, 4 ] Float64Array(3) [ 1, 2, 3 ] false
0.10000000149011612 0.0999755859375
[ 3, 1, 2 ] [ 72, 105 ] Hi
Uint8Array(2) [ 171, 205 ] abcd
```

| Type | Element | Range / precision |
|---|---|---|
| `Int8Array` / `Uint8Array` / `Uint8ClampedArray` | 1 byte | −128…127 / 0…255 (wraps) / 0…255 (clamps and rounds — used by `<canvas>`, the HTML drawing surface, for pixel data) |
| `Int16Array` / `Uint16Array` | 2 bytes | |
| `Int32Array` / `Uint32Array` | 4 bytes | |
| `Float16Array` (ES2025) / `Float32Array` / `Float64Array` | 2 / 4 / 8 bytes | half / single / double precision |
| `BigInt64Array` / `BigUint64Array` | 8 bytes | 64-bit integers as BigInt |

Key differences from normal arrays: **fixed length** (no `push`, `pop`, `splice`), every value is **coerced** to the element type (`256` wraps to `0` in a `Uint8Array`; `-1` becomes `255`), and `Array.isArray` is `false`. Byte order (**endianness**) follows the machine for typed arrays; `DataView` lets you choose it explicitly, which binary file formats and network protocols need.

---

## Chapter TL;DR

- Arrays are ordered, zero-indexed lists — and objects (`typeof` → `'object'`; test with `Array.isArray`). Create with literals; use `Array.from({ length }, fn)` / `Array.of` instead of the quirky `new Array`.
- Read with `arr[i]` or `arr.at(-1)`; missing index → `undefined`. `length` is writable; avoid holes.
- `push`/`pop` are O(1) at the end; `unshift`/`shift` are O(n) at the start. `push` returns the new length.
- **Mutators**: `push pop shift unshift splice sort reverse fill copyWithin`. **Copying twins (ES2023)**: `toSorted toReversed toSpliced with`. Non-mutating: `slice concat map filter flat flatMap` and spread. Never mutate shared state/props.
- `map` transforms, `filter` selects, `reduce` folds (always pass the initial value), `find`/`findLast` return the element, `findIndex` the index, `some`/`every` short-circuit, `includes` finds `NaN`,

`flat` / `flatMap` flatten. `forEach` can't `break` or await.

- Default `sort` is by string; pass `(a, b) => a - b` for numbers, `localeCompare` for text, chain keys with `||`. Sort is stable (ES2019). Shuffle with Fisher–Yates.
- Spread copies shallowly; `structuredClone` deep-copies; `Array(n).fill([])` shares one object. Destructure by position, swap with `[a, b] = [b, a]`.
- Arrays for ordered lists, objects for records, Map/Set for lookups and uniqueness. Typed arrays hold fixed-length binary numbers over an `ArrayBuffer`.